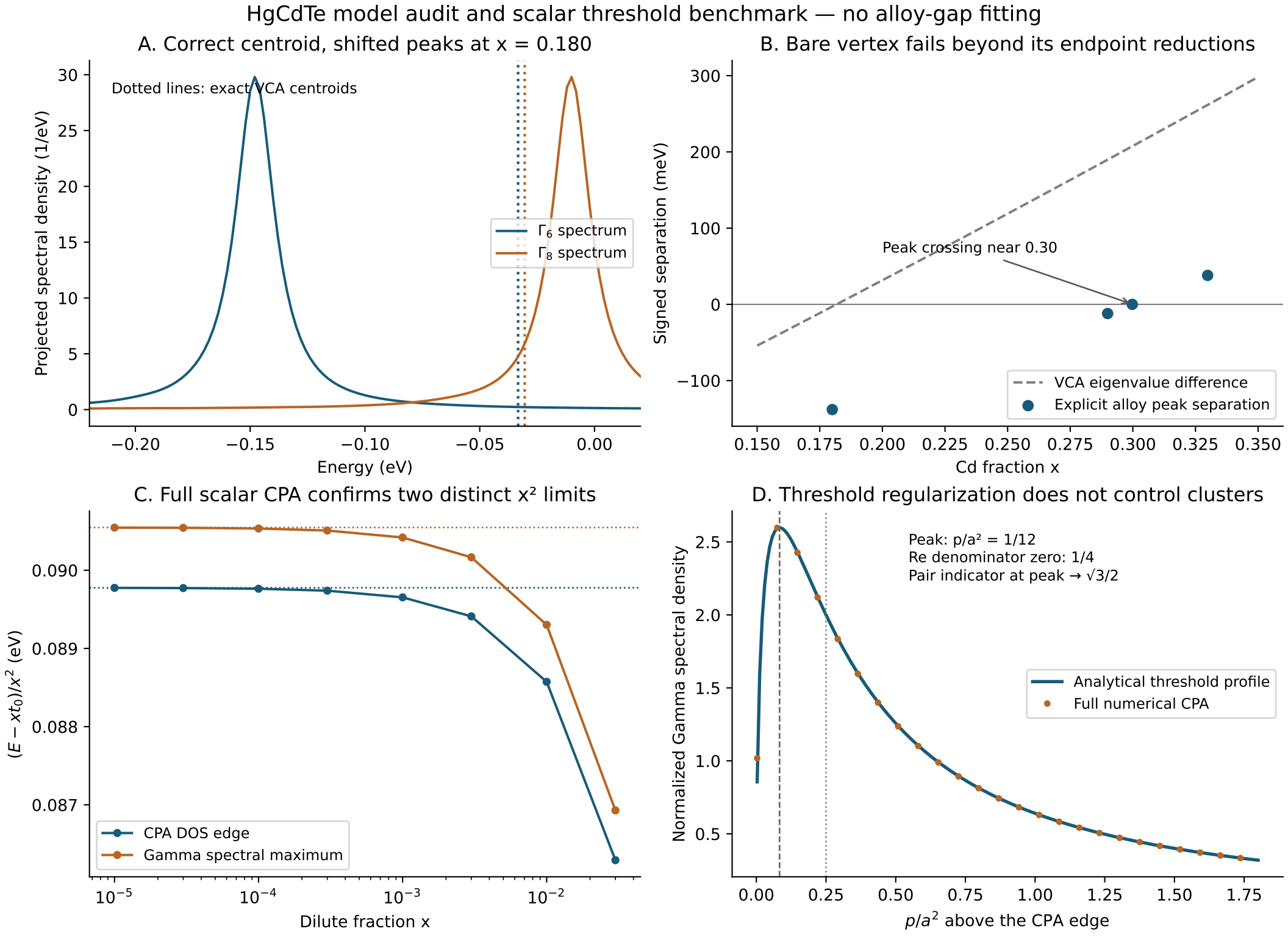
HgCdTe model audit and scalar threshold benchmark — no alloy-gap fitting
A. Correct centroid, shifted peaks at x = 0.180
Dotted lines: exact VCA centroids
Projected spectral density (1/eV)
Energy (eV)
Γ6 spectrum
Γ8 spectrum
B. Bare vertex fails beyond its endpoint reductions
Signed separation (meV)
Peak crossing near 0.30
VCA eigenvalue difference
Explicit alloy peak separation
Cd fraction x
C. Full scalar CPA confirms two distinct x² limits
(E − xt0)/x² (eV)
CPA DOS edge
Gamma spectral maximum
Dilute fraction x
D. Threshold regularization does not control clusters
Peak: p/a² = 1/12
Re denominator zero: 1/4
Pair indicator at peak → √3/2
Normalized Gamma spectral density
Analytical threshold profile
Full numerical CPA
p/a² above the CPA edge